

Complementary and alternative medicine use in rheumatoid arthritis: proposed mechanism of action and efficacy of commonly used modalities

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Abstract Complementary and alternative medicine (CAM) has become popular in patients with Rheumatoid arthritis (RA) worldwide. The objective of this study is to systematically review the proposed mechanisms of action and currently available evidence supporting the efficacy of CAM modalities in relieving signs and symptoms of RA. The prevalence of CAM usage by RA patients is anywhere from 28% to 90%. Many published studies on CAM are based on animal models of RA and there is often insufficient evidence for the efficacy of CAM modalities in RA. The existing evidence suggests that some of the CAM modalities, such as acupuncture, herbal medicines, dietary omega-3 fatty acids, vitamins, and pulsed electromagnetic field show promising efficacy in reducing pain. While the use of CAM modalities for the treatment of RA continues to increase, rigorous clinical trials examining their efficacy are necessary to validate or refute the clinical claims made for CAM therapies.

Keywords Complementary and alternative medicine (CAM) · Alternative medicine · Herbal remedies · Rheumatoid arthritis

Abbreviations

AA Adjuvant arthritis
Bhsp65 Mycobacterial heat shock protein 65
BVA Bee venom acupuncture
CAM Complementary and alternative medicine

COX	Cyclooxygenase
CIA	Collagen induced arthritis
DHA	Docosahexaenoic acid
EA	Electro acupuncture
EGCG	Epigallocatechin-3-gallate
EPA	Eicosapentaenoic acid
GPx	Glutathione peroxidase
GSH	Glutathione
IFN- γ	Interferon gamma
IGF-1	Insulin like growth factor-1
IL-1 β	Interleukin 1 beta
IL-6	Interleukin 6
iNOS	Inducible Nitric oxide synthase
LEW	Lewis
LOX	Lipoxygenase
MDA	Malondialdehyde
MMP	Matrix metalloproteinase
MTX	Methotrexate
NF- κ B	Nuclear factor kappa B
NO	Nitric oxide
PUFA	Polyunsaturated fatty acid
RA	Rheumatoid arthritis
RT-PCR	Reverse transcriptase polymerize chain reaction
SOD	Superoxide dismutase
TCA	Traditional Chinese acupuncture
TNF- α	Tumor necrosis factor alpha
TWHF	<i>Tripterygium wilfordii</i> Hook
VEGF	Vascular endothelial growth factor

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Introduction

According to the National Center for Complementary and Alternative Medicine (NCCAM) “Complementary and alternative medicine is a group of diverse medical and

health care systems, practices, and products that are not presently considered to be part of conventional medicine”.

- The use of complementary and alternative medicine (CAM) has become popular with consumers worldwide. An estimated 40% to 60% of Americans use CAM to manage various medical conditions, and 33% of them cite pain as the primary reason [1]. Arthritis is one of the foremost diseases for which patient seeks CAM options [2, 3]. However, more than 70% of patients using CAM modalities never mention these products to their physicians [4]. Thus, an invisible “mainstream” of alternative care exists in the United States, and little is known by physicians about its safety, efficacy, and mechanism of action.
- Rheumatoid arthritis (RA) affects ~1% of the adult population with women afflicted more frequently than men and has a significant impact on the patient’s physical, emotional, and social functioning [5]. The existing conventional treatments available for RA are no more considered as curative or definitive therapies [6]. Owing to the adverse effects of the medical management, many individuals with RA are turning to CAM modalities for relief of their symptoms. The prevalence of CAM usage by RA patients in the United States is estimated to be anywhere from 28% to 90% [7]. The patients may expe-

rience an adverse outcome from the interaction of CAM modalities with other medications. Therefore, the health-care professionals are feeling the pressure to advice patients, using accurate, scientific information, on their safe use, having had limited, if any, training and reliable sources of information.

- The therapies and practices included within the scope of complementary medicine field can be broadly categorized into: treatments people can administer themselves (e.g., botanical herbs, nutritional supplements, health food, meditation, and magnetic therapy; Table 1), treatments administered by providers (e.g., acupuncture, chiropractic, massage, reflexology, and osteopathic manipulations), and treatments people can administer under the periodic supervision of a provider (e.g., Tai Chi, yoga, Homeopathy, and Ayurveda).

The emphasis of this review is on some of the most widely used CAM modalities with promising evidence of efficacy in reducing the signs and symptoms of RA, by elucidating the scientific basis of their anti-inflammatory and/or immunomodulatory activities. An understanding of the pathophysiology of CAM modalities could allow physicians to counsel effectively on their proper and improper use, prevent adverse drug interactions, and anticipate toxicities.

Table 1 Proposed mechanisms for anti-inflammatory effects of herbal medicines in RA

CAM modality	Active component	Mechanism of action
1. <i>Camellia sinensis</i> (green tea)	Epigallocatechin-3-gallate (EGCG)	Inhibits TNF- α , IFN- γ , NF- κ B, iNOS, and COX-2
2. <i>Celastrus aculeatus</i>	Sesquiterpene esters (celastrol, celaphanol) and flavonoids (epiafzelechin)	Inhibits iNOS (NO production) Facilitates anti-inflammatory IL-10 production
3. <i>Uncaria</i> (cat’s claw)	Pentacyclic oxindole alkaloids	Inhibits TNF- α , NF- κ B, iNOS, IL-1 β -induced NO production Promotes anabolic IGF-1 production
4. <i>Lepidium meyenii</i> (RNI 249)	Not specified	Autocrine activation of IGF-1
5. <i>Tripterygium wilfordii</i> Hook F (Thunder God Vine)	Triptolide	Inhibits IL-1, IL-17, TNF- α , LPS-induced MMPs production, and iNOS expression
6. <i>Perna canaliculus</i>	Not specified	Inhibits TNF- α , IL-12p40, superoxide release and arachidonic acid cascade (COX-2)
7. <i>Curcuma longa</i> (turmeric)	Curcumin (diferuloylmethane)	Inhibits NF- κ B, MMP gene expression, NO production, arachidonic acid cascade (phospholipases A2, C γ 1)
8. <i>Curcuma phaeocaulis</i>	Not specified	Inhibits cytokine pathways, COX-2, LPS-induced iNOS expression
9. <i>Zingiber officinale</i> (ginger)	[6]-Gingerol	Inhibits LPS-induced NO production, COX and LOX pathways (PGE-2, LTB4 production), and peroxynitrite-mediated damage
10. <i>Semecarpus anacardium</i> Linn.	Flavonoids	Inhibits TNF- α , NO, phospholipase A2, oxidative stress (ROS), and modulates T-cell and B-cell immune responses

Methods

The key terms, *complementary medicine*, *alternative medicine*, were used to search MEDLINE and PUBMED (1988–2008) and cross-referenced with the key term *rheumatoid arthritis* for all the available full-text articles in the English language. Studies identified by the search strategies were assessed for inclusion in the review. An initial screen of the identified studies excluded articles the titles and/or abstracts of which were clearly irrelevant. The modalities with currently available scientific evidence regarding the efficacy and mechanism of action were included with a particular emphasis to include available randomized controlled trials (RCTs). While the emphasis was on human studies, a few selected animal studies were included which provided important clues about the modalities' possible mechanism of action.

Results

CAM modalities in RA

Acupuncture

Acupuncture has gained popularity in the United States with an estimated use by 1 million Americans annually, primarily for pain relief [8]. The Traditional Chinese model is based on the hypothesis that a force of energy (Q_i) is circulating in the body in some channels or meridians. If the energy is able to flow freely in the body, the patient is healthy and without pain, and if for various reasons, this flow gets blocked, the patient will be ill or in pain. Insertion of acupuncture needles in various well-defined points restores the free flow of energy and the patient is cured (Table 2).

The Western scientific model is based on the “gate control” theory, whereas acupuncture releases neurotransmitters which act at the second layer of the dorsal horn that can effectively block any incoming pain transmission via the C-fibers, by releasing enkephalin [9]. These substances cause analgesia by decreasing pain and modulating stress. Although there is clear distinction between the traditional and western acupuncture, however, the approaches

considerably overlap [10] as the acupuncture points are identified by their location in the meridians.

- Acupuncture involves various techniques for manual stimulation of needles, such as electrical stimulation and heat stimulation of needles, which is known as moxibustion, wherein the moxa herb, *Artemisia vulgaris* is burned at the handle end of the needle. Injection acupuncture, involving the injection of herbal extracts into acupuncture points, is occasionally used [11].
- A recent controlled clinical study conducted by Zanette et al. [12], evaluated the efficacy of acupuncture as an adjunctive therapy in RA. As much as 40 RA patients were recruited after matching the baseline characteristics, and were randomized to receive either a standard protocol of acupuncture (AC) or superficial acupuncture at non-acupuncture points (control AC) for a period of 9 weeks. The assessment was made after 5 and 10 treatment sessions and after 1 month of follow-up. The primary outcome was assessed through achievement of ACR 20% criteria, and the secondary measures of assessment included disease assessment scale (DAS), tender and swollen joint count, morning stiffness, Health Assessment Questionnaire (HAQ), visual analog scale (VAS) of pain, physician global assessment of activity disease, physician and patient global assessment of treatment, and inflammatory markers (Erythrocyte Sedimentation Rate and C-reactive protein). A trend for better efficacy was noticed in the AC group, while assessment during their 4th visit/last visit/follow-up visit (i.e., 1 month after 10th session), as 40% patients of AC group achieved ACR 20 improvement criteria in comparison to 10% patients of control AC group, but this difference did not reach statistical significance ($p = 0.068$). Among the secondary outcomes, a statistically significant improvement was observed in the physician's global assessment of disease activity ($p < 0.001$), during the assessment at the 3rd visit (i.e., after 10th session). In addition, a favorable trend was seen within the AC group in most of the variables (HAQ, VAS for pain, morning stiffness, and patient and physician's global assessment of treatment) in relation to the baseline characteristics.

Table 2 Proposed mechanisms for beneficial effects of other CAM modalities in RA

CAM modality	Mechanism of action
1. Acupuncture	Blockade of pain pathways (C-fibers) by enkephalin release
2. Electro-acupuncture	Suppression of IL-6, TNF- α , IFN- γ , T cells, and B cells
3. Bee-venom Acupuncture	Counter-irritation method (noxious stimuli increasing pain threshold)
4. Omega-3 PUFAs (fish oil)	Produce anti-inflammatory lipids (resolvins, protectins) which inhibit IL-1 β , TNF- α , and NF- κ B
5. Antioxidant vitamins	Improvement of the antioxidant parameters (glutathione, total thiols, and vitamin C) and reduction of the oxidative stress marker (malondialdehyde)

- Another double-blind controlled pilot study recruiting 36 patients was performed to compare the effects of Traditional Chinese acupuncture (TCA), Electro acupuncture (EA) and Sham acupuncture (Sham) in patients with RA [13]. The clinical methods of assessment of pain included visual analog scale (VAS) of pain, the number of swollen and tender joints, disease activity score using the 28 joint counts (DAS 28), and the Health Assessment Questionnaire (HAQ). Although the pain scores did not differ between the three groups, however, this study demonstrated a significant reduction in the number of tender joints in patients with RA after acupuncture treatment.
- To observe the influence of acupuncture on RA, electro-acupuncture (EA) was used in a study [14], to objectify the amount of acupuncture stimulation on an experimental model of collagen-induced arthritis (CIA) in a mouse. CIA model replicated the human rheumatoid arthritis and was prepared by immunization of the susceptible mice with type II collagen (CII) [15]. Electro acupuncture given at ST-36 (stomach) meridian with electrical current at a frequency of 2 Hz showed significant reduction in the inflammatory mediators, such as IL-6, TNF- α , IFN- γ , collagen II antibody, and IgG and IgM levels in CIA mice serum and a reduced CD11b+/Gr1+ cell counts in CIA mouse population, indicating reduced inflammatory granulocytes [16–20].
- Electro acupuncture at ST36 meridian also caused significant reduction of CD69+/CD3e+ cells and CD11a+/CD19+ cells in lymph nodes of mouse population, indicating suppressive effects on T-cell and B-cell activation in CIA [21–23]. T cells, B cells, and macrophages are the major causes involved in the pathogenesis of RA [21, 23–25]. Therefore, it was speculated that EA at ST36 may have an anti-inflammatory and anti-arthritic effect on CIA by suppressing the autoimmunity and modulating the immune abnormality.
- Observations suggest that acupuncture without electrical stimulation showed a similar tendency but weaker effect than that of Electro-acupuncture on CIA. Hence there may be an involvement of neuro-immune interaction in the mechanism of acupuncture on immune diseases like RA [14].
- Besides all the supportive evidence, a clinical trial by David et al. [26] refuted acupuncture as an effective adjunct in the treatment of RA. The reason behind this refute was believed to be the enrollment of patients with established erosive RA, which could be less responsive to acupuncture as compared to early pre-erosive, more inflammatory disease.
- In summation, even though no clear anti-inflammatory effects have been demonstrated, acupuncture may be, in some cases, a useful adjunct measure in reducing pain in RA.

Bee venom acupuncture

Bee venom acupuncture (BVA) exerts dual action, a pharmacological action from the bioactive compounds isolated from bee venom and mechanical action from the acupuncture stimulation. BVA has a good evidence for its effectiveness in pain syndrome, herniation nucleus pulposus, cervical disc protrusion, and progressive muscle atrophy [27].

Anti-inflammatory and anti-nociceptive effects of BVA have been demonstrated in various animal arthritic models:

- Direct injection of bee venom into acupoint ST36 produced a potent anti-nociceptive effect in an adjuvant-induced RA animal model [27, 28].
- BVA at acupoint BL23 significantly reduced proteolytic enzyme activities and level of reactive oxygen species (ROS)-induced oxidative damage to synovial fluid proteins in a type II collagen-induced arthritis in rats.
- BVA at acupoint GB34 on lipopolysaccharide (LPS)-induced arthritis demonstrated decrease in numbers of white blood cells, infiltration of leukocytes, and fibroblasts into synovial joints, CD56, IL-1b, IL-2R, CD54, and CD106 into synovial membrane.

In conclusion, the effects of BVA depend upon the locations injected. BVA at acupoints exert stronger effects than non-acupoints [27].

- A randomized controlled trial by Lee et al. [27, Table 3], and two uncontrolled clinical trials by Lee et al. and Kwon et al., respectively, were conducted to study the effects of BVA on RA patients [27]. The results showed a statistically significant decrease in tender joint count, swollen joint count, morning stiffness, ESR, CRP levels, and HAQ scores after treatment with BVA.
- Safety intervention is important pertaining to serious adverse effects of BVA, such as anaphylaxis and non-serious adverse effects, such as local swelling or itching, but is infrequently provided that it is practiced according to established safety rules in appropriate anatomic regions. However, it would be worth considering BVA as an applicable adjunctive treatment for RA.

Herbal medicines for RA

Herbal medicines described in traditional texts, have been used as anti-inflammatory, anti-microbial, and anti-viral medicine for the cure of RA, allergies, infections, wound healing, and fever [29]. Because of the limited evidence about the long-term effects of currently available herbal medicines and its associated side effects, the American College of Rheumatology recommends the careful use of dietary supplements and herbal medicines during early

stages of treatment in order to limit the degree of joint destruction.

Joint inflammation and cartilage loss is the hallmark of arthritis which primarily involves the activation of catabolic pathways and inhibition of anabolic pathways.

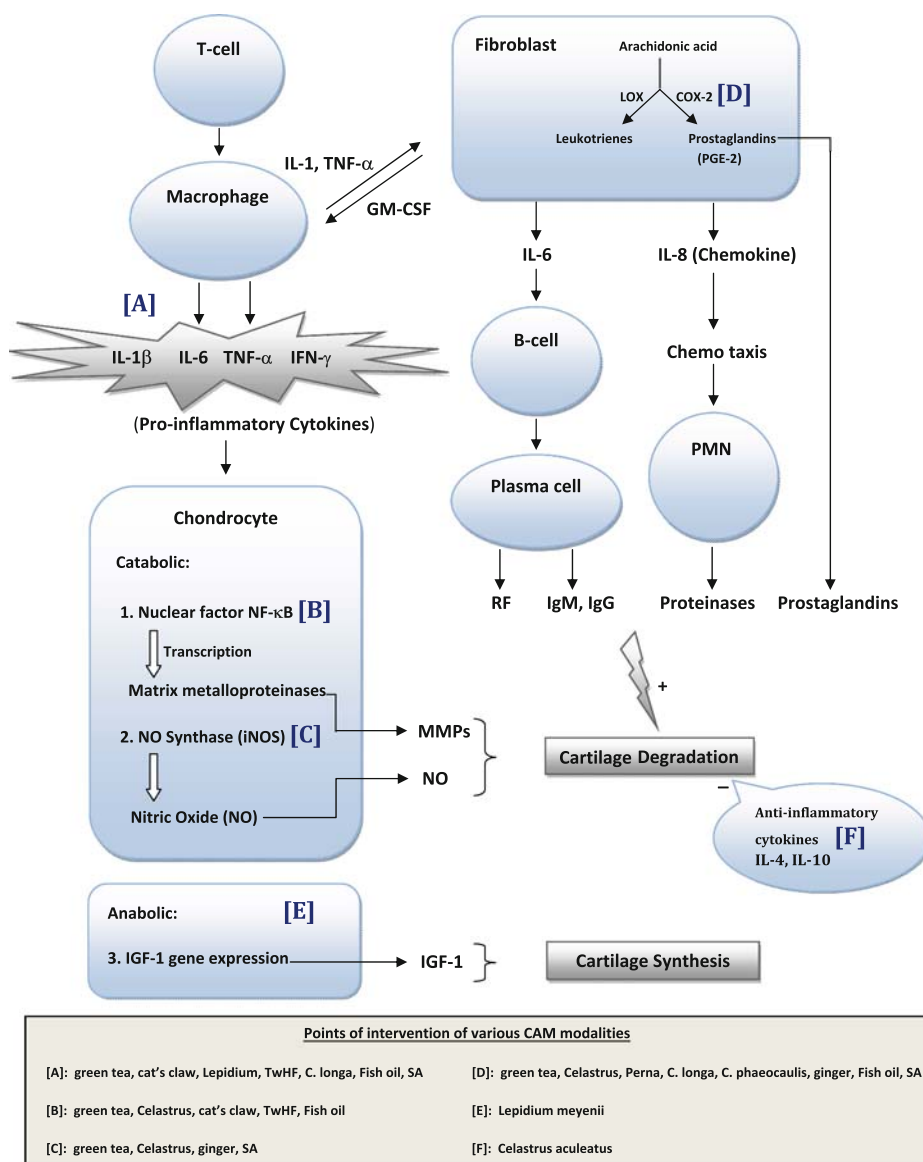
- The Catabolic process is mediated by pro-inflammatory cytokines like IL-1 β , IL-6, TNF- α , IFN- γ , and NF- κ B which induce the matrix metalloproteinases (MMPs) on human cartilage causing extra cellular matrix degradation [30–32].
- The Anabolic process is mediated by insulin-like growth factor (IGF-1).

The pharmacological basis and scientific evidence regarding the efficacy and side effects of the most commonly used herbs for the treatment of RA is given in Fig. 1.

Camellia sinensis (green tea) Green tea is one of the most commonly consumed beverages in the world with no reported side effects. The pharmacological properties of green tea are attributed to its high content of polyphenols/catechins, mainly epigallocatechin-3-gallate (EGCG). Most of the effects of green tea are attributed to its constituent polyphenol EGCG [33, 34].

- Studies revealed that EGCG inhibits the transcription factor nuclear factor kappa-B (NF- κ B) in conjunction with IL-1 β inducible nitric oxide synthase (iNOS) and cyclooxygenase (COX-2), resulting in reduction of nitric oxide and prostaglandin E2 (PGE2) in vitro [35, 36].
- EGCG has also been shown to selectively inhibit the IL-1 β -induced phosphorylation of c-Jun-N-terminal

Fig. 1 Targets of CAM modalities in the inflammatory cascade in RA



kinase (JNK), which is primarily responsible for inflammatory and degenerative diseases [37, 38].

- Among the matrix metalloproteinases (MMP), MMP1 and MMP13 levels are found to be significantly elevated in the RA joints. Studies revealed that pretreatment of human chondrocytes with EGCG, inhibited the expression and activities of MMP1 and MMP13 in vitro [39].

The catechins constituent of green tea were shown to inhibit the degradation of human cartilage proteoglycan and type II collagen, and selectively inhibit the aggrecanases called disintegrating and metalloproteinase with thrombospondin motifs (ADAMTS)-1, -4 and -5 [40, 41].

- The effects of green tea were demonstrated in an animal model of inflammatory polyarthritis, wherein the collagen-induced arthritis (CIA) was ameliorated by prophylactic administration of green tea polyphenols (GTPs) in drinking water [42]. The reduction in the incidence and severity of CIA was believed to be due to marked inhibition of the inflammatory mediators COX2, IFN- γ , and TNF- α in the joints of green tea fed mice. In addition, the total immunoglobulins (IgG) and type II collagen specific IgG levels were found to be lower in the serum and arthritic joints of GTP fed mice. The reduction in the inflammatory mediators correlated with marked reduction of inflammation in the synovium seen on histopathology.

Adding up to the evidence of anti-inflammatory effects, the global availability of green tea, low cost, and proven lack of toxicity may prove it to be an effective adjunct in the treatment of RA.

Celastrus aculeatus Merr. (*Celastrus*) *Celastrus* is a medicinal herb of Chinese origin which has been used for centuries to treat RA, osteoarthritis (OA), and lower-back pain. *Celastrus* and some of its constituents namely, *sesquiterpene esters* and *flavonoids* have been shown to possess anti-inflammatory, anti-oxidant, and anti-cancer properties [43–48]. The proposed mechanisms for the anti-inflammatory effects of *Celastrus* include the inhibition of NF- κ B, Inos, and COX-mediated pathways [45–49].

- A recent study performed using the adjuvant arthritis (AA) model examined the effects of *Celastrus* on the T-cell and antibody responses to the disease-related antigen [50]. The results revealed that *Celastrus* enhanced the production of protective antibodies during the course of AA [51, 52]. It has been proposed that the protective effect of antibodies in AA is probably mediated by facilitating the secretion of anti-inflammatory cytokine IL-10 from mononuclear cells [51]. The effects of *Celastrus* on AA were comparable to that of MTX, not only in

preventing the induction of arthritis but also halting the progression of ongoing arthritis.

Cat's Claw (*U. tomentosa*, *U. guianensis*, *vincaria*^R) *Cat's claw* is a Peruvian vine with well-documented medicinal properties in alternative medicine literature. *Cat's claw* extract has been shown to possess antioxidant, anti-inflammatory and immunomodulating properties [53, 54] owing to its inhibitory action on NF- κ B, TNF- α and associated inflammatory mediators [54–57].

- The chemical composition of the aqueous extract of *U. tomentosa* includes pentacyclic oxindole alkaloids (absent in *U. guianensis*), tannins, quinovic acid, glycosides, falconoid and sterols [58]. Pentacyclic oxindole alkaloids are the most investigated of all the constituents in *U. tomentosa* extract, for anti-inflammatory and immunomodulating, which are reported to induce a yet unknown immune regulating factor [53].
- A two phase, double-blind, randomized clinical trial (Table 3) demonstrated that an extract from *U. tomentosa*, that is rich in pentacyclic oxindole alkaloids, showed a reduction in the number of painful joints as compared to placebo among RA patients (53.2% vs. 24.1%; $p = 0.044$) [59]. The patients also experienced a statistically significant improvement in the number of painful ($p = 0.003$) and swollen joints ($p = 0.007$) and the Ritchie index ($p = 0.004$) compared to the values after 24 weeks of placebo.
- However, it is speculated that the antioxidant and anti-inflammatory effects of the extracts of *U. tomentosa* and *U. guianensis* may be independent of their alkaloid content, because several assays showed that the extract of *U. guianensis* (despite the absence of pentacyclic alkaloids) was more potent in scavenging free radicals and inhibiting TNF- α production in mouse macrophages [55].
- *Cat's claw* extract have also shown to effectively inhibit the lipopolysaccharide (LPS)-induced free radical production and subsequent lipid peroxidation in macrophage cell lines [57] in addition to inhibiting TNF- α production and NF- κ B expression.
- Several clinical studies and in vitro bioassays on the safety and pharmacological profile of *cat's claw* have documented non-toxic nature of *cat's claw* [59–61]. *Cat's claw* extract has successfully completed a small trial for osteoarthritis of knee demonstrating a rapid reduction in pain and improved function [60].

Lepidium meyenii (RNI 249) A natural product of South American origin has shown a potential ability not only to limit the joint inflammation and destruction but to promote joint restoration also. *Lepidium meyenii* (RNI 249) possesses

Table 3 Positive clinical studies supporting the efficacy of widely used cam modalities in RA

CAM modality	Clinical studies (First author)	Number of RA patients enrolled	Duration of study	Significant outcome measure and results ($p < 0.05$)
1. Acupuncture	Zanette [12]	40	5 weeks	PGA
	Tam [13]	36	10 weeks	TJC, PGA
2. Bee venom acupuncture	Lee [27]	80	2 months	TJC, SJC, MS, ESR and CRP
3. <i>Uncaria tomentosa</i> (cat's claw)	Mur [59]	40	52 weeks	TJC, SJC, RI
4. <i>Tripterygium wilfordii</i> hook F	Tao [72]	35	20 weeks	ACR 20% response: high dose > low dose
5. <i>Curcuma longa</i> (turmeric)	Deodhar [90]	18	4 weeks	MS, joint swelling, and walking time
6. Omega-3 PUFAs (Fish oil)	Galarraga [113]	97	9 months	Relative reduction in the NSAIDs requirement by >30%
	Berbert [114]	43	Assessment after 12 and 24 weeks	Improvement in MS, jt. pain intensity, RI, and handgrip strength
	Sundrarjun [128]	60	24 weeks	ACR 20 in 2 pts. (fish oil, placebo group); Reduced levels of IL-6, TNF- α , sTNF-R p55, and CRP
7. PEMF	Segal [126]	64	1 week	Pain reduction, S-GADA

ACR 20 American College of Rheumatology 20% improvement criteria, TJC tender joint count, SJC swollen joint count, MS morning stiffness, JS joint score, PGA physician global assessment, RI Ritchie index, ESR erythrocyte sedimentation rate, CRP C-reactive protein, S-GADA subject's global assessment of disease activity, TNF tumor necrosis factor, sTNF-R p55 soluble tumor necrosis factor receptor p55; IL-6 Interleukin 6

an uncertain mechanism for treating arthritis. Interestingly, RNI 249 has well-defined applications for fertility and vitality [62–64]. However, based on the observational studies that these extracts were anabolic in farmed fish (unpublished reports) the growth of which is critically dependent on IGF-1, the mechanism of action of RNI 249 was hypothesized as autocrine activation of IGF-1.

- The effects of RNI 249 extract evaluated in primary cultures of human chondrocytes revealed significant enhancement in the expression of IGF-1 gene; however, when RNI 249 and cat's claw extracts were combined, the effects were additive resulting in a nearly 4-fold increase in IGF-1 gene expression [65].
- Furthermore, RNI 249 and cat's claw extract protected the IGF-1 production by chondrocytes to normal levels even in the presence of pro-inflammatory cytokines like IL-1 β and TNF- α in the culture medium, which are expected to completely suppress the production of IGF-1 β by chondrocytes [32, 66, 67].
- The combination of RNI 249 and cat's claw extract has already been commercialized in the USA and Canada and marketed as a pain-reducing supplement for patients with arthritis [65].

Tripterygium wilfordii Hook F TwHF also known as 'Thunder God Vine' is a perennial plant grown in China and Taiwan. The medicinal extract, derived from the roots of TwHF plant, has been used in treating various autoimmune and inflammatory diseases, such as RA, SLE, psoriasis, and asthma [68].

- The biologically active component believed to be responsible for the anti-inflammatory effects of TwHF is Triptolide [69]. Various animal studies have shown that Triptolide suppresses the IL-1 and LPS-induced production of pro matrix metalloproteinases MMP-1 and MMP-3 in cultured synovial fibroblasts and mouse macrophages [70]. A study has also revealed an inhibitory effect of triptolide on IL-1, IL-17, and TNF- α -induced expression of aggrecanases gene in human chondrocytes [71]. Triptolide has also demonstrated an inhibitory action of an iNOS gene expression by downregulating NF- κ B DNA-binding activity and c-Jun-N-terminal kinase (JNK) pathway.
- A double-blind, placebo-controlled study has been reported showing the beneficial effects of the ethyl acetate extracts of TwHF in treatment-refractory RA patients [72]. Two groups of patients received ethyl acetate extracts of TwHF in doses of 180 and 360 mg/day, respectively, for 20 weeks. Results have shown that the patients treated with high-dose TwHF achieved rapid ACR 20% improvement as compared to placebo ($p = 0.0001$). Whereas, the efficacy of low-dose TwHF was less than that of high-dose extract ($p = 0.027$) but was still greater than that of placebo ($p = 0.0287$). Moreover, 50% of patients on high-dose TwHF, and 10% of patients on low-dose TwHF achieved ACR 50% improvement response. However, both treatment groups showed a significant decrease in the number of tender joints and an improvement in physician's global assessment.

- Another study involving administration of the extracts of TwHF in doses >360 mg/day to the RA patients also showed a significant improvement in the clinical manifestations and laboratory parameters [73].
- The above studies reflected that the ethyl extract of TwHF at dosage as high as 570 mg/day appeared to be safe and doses >360 mg/day were associated with clinical benefits in RA patients.
- Both the studies showed that the TwHF extract was well tolerated by the patients and no toxic effects other than diarrhea were observed in the patients receiving the highest dose. However, a few other studies have reported that the most common adverse effects of TwHF include vomiting, diarrhea, abdominal pain, hair loss, and amenorrhea [74].
- Therefore, while the TwHF extract has shown some efficacy in RA, the vigilant physician should be aware of potential toxicity associated with the agent.

Perna canaliculus (Perna) Perna, a New Zealand green-lipped mussel, has demonstrated potent anti-inflammatory properties, comparable to that of non-steroidal anti-inflammatory drugs (NSAIDs) [75]. Perna mussel was for centuries a significant part of the traditional diet of the Maori inhabitants of New Zealand. The reported incidence of arthritic and rheumatic disorders was extremely low among coastal New Zealanders compared to those living inland [76].

Experiments performed to evaluate the effects of Perna have revealed both the humoral and cellular immunomodulatory effects of Perna [77]. A study on AA model attributed anti-arthritic effects of Perna to the presence of omega-3 PUFAs, which inhibits the biosynthesis of proinflammatory prostaglandins (PGE₂), by blocking the arachidonic acid metabolism through COX pathways [78]. Reports have also shown that green-lipped mussel extract increased the gestation period of rats, suggesting that Perna contains active inhibitors of prostaglandin production [79].

Another animal study demonstrated the beneficial effects of Perna due to the reductions in anti-collagen antibody levels, pro-inflammatory cytokine production (TNF- α and IL-12p40), superoxide release, and amelioration of established CIA [80].

It has been suggested that, unlike NSAIDs, the green-lipped mussel preferentially blocks COX-2 enzyme activity in relation to COX-1, offering potential GI protection to ulcer-prone patients.

Curcuma longa (Turmeric) Turmeric is a commonly used coloring/flavoring agent with a history of its use in Ayurveda for various medicinal conditions [81–83].

- The major component of turmeric is curcumin (diferuloylmethane) which constitutes approximately 90% of

total curcuminoid content. Curcumin is a potent inhibitor of the common transcription factor NF- κ B [84–86]. Studies have also shown the inhibitory effect of curcumin on the arachidonic acid cascade (COX-2 and LOX) by inhibiting the catalytic activities of phospholipases A₂, C γ 1, and D [87–89]. Curcumin also blocks the catabolic effects of IL-1 β -induced upregulation of MMP-3, and IL-1 β -induced decrease in type II collagen synthesis, that are known contributors in the pathogenesis of RA.

- A double-blind crossover clinical trial (Table 3) was conducted, enrolling 18 RA patients, in which the administration of curcumin (1,200 mg/day) for 2 weeks was followed by phenylbutazone (300 mg/day) for another 2 weeks. The patients reported significant improvement in morning stiffness, joint swelling, and walking time [74, 90].

Curcuma phaeocaulis A study by Tohda [91] involving comparison of anti-inflammatory activities of six species of *Curcuma* rhizomes, demonstrated that the methanol extract of *Curcuma phaeocaulis* significantly reduced the paw swelling and the inflammatory marker, haptoglobin in serum of arthritis model mice, whereas other *Curcuma* species (*Curcuma longa*, *Curcuma wenyujin*, *Curcuma kwangsiensis*, *Curcuma zedoaria*, and *Curcuma aromatica*) had no clear effects on inflammation.

The anti-inflammatory effects of *Curcuma phaeocaulis* are believed to be due to inhibition of COX-2 expression and activation [89]. Despite the fact that *Curcuma phaeocaulis* contains very low amounts of curcuminoids, it showed anti-inflammatory activity, suggesting that a curcuminoid-independent pathway may exist for its anti-inflammatory activity.

Zingiber officinale (Ginger) Ginger is a very commonly used dietary constituent worldwide, and is known to possess antioxidant, anti-inflammatory, antiseptic, and carminative properties [92]. Ginger has a history of its use in Ayurveda for treating inflammatory and rheumatic diseases.

- The anti-inflammatory effects of ginger in treating arthritis are believed to be due to [6]-gingerol, which is a pungent phenolic constituent of ginger. [6]-Gingerol inhibits the LPS-induced NO production and effectively protects against the peroxynitrite-mediated damage [93]. Studies have shown that gingerols were excellent inhibitors of LPS-induced PGE₂ production [94]. In conclusion, the beneficial effects of ginger were attributed to the inhibition of COX and LOX cascade-mediated PGE₂ and LTB₄ production in the affected joints [87, 95].
- Srivastava et al. [95, 96] has reported case histories supporting the efficacy of ginger in patients with RA, OA,

and muscular discomfort. Overall, the conclusions reflected the voluntary response of 56 patients (28 with RA, 18 with OA, and 10 with muscular discomfort), regarding the effect of ginger consumption on the usual symptoms—pain and swelling—in arthritis. Observations suggest that three quarters of arthritis patients, and all the patients with muscular discomfort experienced marked reduction in pain after consumption of ginger.

- Another limited study performed on seven RA patients, reported the beneficial effects of ginger in alleviating the pain and swelling of the involved joints [95]. Though many clinical studies have shown statistically significant effects of ginger in alleviating the symptoms of OA [97, 98], large-scale controlled clinical studies are essential to prove the beneficial effects of ginger in RA patients.

Semecarpus anacardium Linn. *Nut milk extract* *Semecarpus anacardium* is a deciduous tree distributed in the sub-Himalayan tract and in tropical parts of India. In traditional medicine, it is highly valued for the treatment of gout, rheumatic pains, and cancer [99]. The chemical constituents of the milk extract of *Semecarpus anacardium* Linn. includes flavonoids, phenols, and carbohydrates [100]. Reports have shown that *Semecarpus anacardium* Linn. nut milk extract (SA) at a dose of 150 mg kg⁻¹ body weight, is effective against adjuvant arthritis [101]. Flavonoids and polyphenols are considered responsible for the antioxidant and anti-inflammatory effects of SA.

- The protective antioxidant role of flavonoids, is shown to be due to their inhibitory effects on the production of reactive oxygen species (ROS) by their free-radical quenching activity, and their potential to improve the levels of antioxidants in the body [102]. Flavonoids have also been reported to exhibit anti-inflammatory activity by inhibition of phospholipase A2, thereby reducing the production of proinflammatory PGE2, and also by reducing the elevated levels of TNF- α and NO [99].
- Another protective effect of SA is by enhancing the stability of lysosomal membrane, thus preventing the rupture and release of lysosomal enzymes [103], which play a major role in erosive synovitis in RA.
- Furthermore, SA has been shown to modulate both the humoral and cell-mediated immune responses, along with its anti-inflammatory effects in adjuvant arthritic models [100]. The humoral immunomodulatory response was explained by reversion of the elevated levels of IgG and IgA, and the cell-mediated immunomodulatory response by inhibition of T-lymphocyte migration to the inflamed joints.
- Overall, several observations have supported SA's potent anti-inflammatory and anti-arthritic properties that should be substantiated by careful clinical studies.

Herbs necessitating evaluation in RA

The rapidly expanding spectrum of herbal supplements by RA patients has necessitated the evaluation of any possible anti-inflammatory or anti-arthritic properties, often employing in vitro models that use human chondrocytes/cartilage explants or in clinical studies. Many herbal therapies with potential anti-inflammatory or immunomodulatory properties, as reported in other inflammatory conditions, have still not been studied in RA. Furthermore, the following two herbal supplements, Devil's claw (*Harpagophytum Radix*) and Dandelion (*Taraxacum officinale*) carry some evidence of anti-arthritic properties in osteoarthritis and their proof of efficacy in RA is pending.

- Devil's claw (*Harpagophytum Radix*) has been extensively evaluated with RCT's in OA patients. Although clear mechanism has not been established but the probable explanation for anti-arthritic effects of *harpagophytum* include the ability to suppress the pro-inflammatory mediators (PGE2, TXB2, IL-1 β , and TNF- α). As suggested in a review by Denner et al. devil's claw extract is most effective at daily dosage of at least 50 mg [104].
- Dandelion (*Taraxacum officinale*) has a history for its use in treating dyspepsia, gallstones, and arthritis. The principle constituents responsible for dandelion's effects include sesquiterpene lactones, previously referred to as taraxacin. A recent study by Jeon et al. [105] demonstrated the anti-angiogenic, anti-inflammatory, and antinociceptive activities of dandelion through its inhibition of NO production, COX-2 expression, and anti-oxidative property by scavenging action on reactive oxygen species (ROS).

Omega-3 polyunsaturated fatty acids and fish oil

The putative role of omega-3 PUFAs in reducing the inflammation and alleviating the symptoms of RA have been a topic of continuous interest. Fish oil is a good source of long chain ω -3 PUFAs, containing 18% EPA (eicosapentaenoic acid) and 12% DHA (docosahexaenoic acid) derived from marine fish [106]. Omega-3 fatty acids are converted by the liver into EPA and DHA. The anti-inflammatory effects of long chain ω -3 PUFAs, EPA, and DHA appears to be due to the production of anti-inflammatory lipids (i.e., resolvins and protectins) which inhibit the activation of transcription factor NF- κ B, and the release of IL-1 β and TNF- α , the key regulators of inflammation [107–110].

- Long chain ω -3 PUFAs also compete with other ω -6 fatty acids (e.g., linoleic acid) for incorporation into phospholipids. The ω -6 fatty acids are metabolized to

arachidonic acid and the inflammatory eicosanoids (i.e., prostaglandins, leukotrienes, and thromboxanes) thus playing pro-inflammatory role. Therefore, it has been suggested that a reduction in the intake of ω -6 fatty acids would increase the effectiveness of ω -3 PUFAs. A study on RA patients demonstrated greater efficacy of ω -3 PUFAs with consumption of low ω -6 PUFAs/arachidonic acid diet [111].

- A meta-analysis of 17 randomized controlled trials was conducted with Cochrane Review Manager 4.2.8 to assess the analgesic effects of omega-3 PUFAs in patients with RA [112]. The results reported that supplementation with high-dose ω -3 PUFAs or fish oil (at least 2.7 g/day of EPA and DHA) for 3–4 months caused a significant reduction in the intensity of joint pain, morning stiffness, and the number of tender joints.
- Recently a double-blind, randomized controlled trial (Table 3) was conducted by Galarraga [113] with an objective to determine whether cod liver oil (ω -3 PUFAs) supplementation helps reducing daily NSAIDs requirement. Results showed that, 39% patients of cod liver oil group and 10% patients of placebo group were able to reduce their daily NSAIDs requirement by >30% ($p = 0.002$), after 9 months of supplementation.
- Another clinical trial by Berbert [114] assigned 43 RA patients to one of three groups, placebo (G1), fish oil (G2), or fish oil plus olive oil (G3). The patients were evaluated after 12 and 24 weeks of treatment. Results revealed statistically significant improvement in groups G2 and G3 in comparison to G1 with respect to joint pain intensity, handgrip strength, duration of morning stiffness, onset of fatigue, and Ritchie's index (Table 3).

Therefore, dietary supplements with long chain ω -3 PUFAs, eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) may be an effective adjunct in the treatment of RA.

Antioxidant vitamins

The scavenging action of oxygen-free radicals have been implicated as a mediator of tissue damage and inflammation perpetuating process in patients with RA [115]. Oxygen-free radicals are responsible for the process of lipid peroxidation, which involves oxidative deterioration of unsaturated fatty acids [116]. Lipid peroxidation affects adversely the membrane-associated polyunsaturated fatty acids and protein thiols [117]. In healthy individuals, where the production of reactive oxygen species (ROS) is low, lipid peroxidation is inhibited by various antioxidants present in the plasma. Contrarily, the patients with rheumatoid arthritis are exposed to oxidative stress and are more prone to lipid peroxidation. Overwhelming oxidative stress and

failure of the antioxidant defense mechanisms are responsible for lipid peroxidation and tissue damage in rheumatoid arthritis [118].

- A study by Jaswal [118] aimed at evaluating the importance of antioxidant therapy in managing RA patients. The study included 40 patients of RA, further randomly divided into two subgroups of 20 each. One group received the conventional treatment for 12 weeks and the other group received antioxidants in the form of vitamins A, C, and E along with the conventional treatment for the same period. Blood samples collected at the time of presentation and after 12 weeks of treatment, were analyzed for concentrations of total thiols, glutathione, vitamin C, and malondialdehyde (MDA).
- The concentrations of total thiols, glutathione, and vitamin C served as antioxidant parameters, whereas malondialdehyde (MDA) levels served as a marker of oxidative stress. Disease activity of patients was also measured at the time of presentation and after 12 weeks of treatment, using the rheumatoid arthritis disease activity index (RADAI).
- The concentrations of total thiols, glutathione, and vitamin C were found to be significantly lower in patients with RA, as compared to healthy controls. A statistically significant increase was found in the post-treatment concentrations of these antioxidant parameters.
- Conversely, the concentrations of MDA and RADAI score were found to be much higher in patients with RA, as compared to normal controls. The evidence of increased concentrations of MDA in patients of RA, is in line with previous reports [119–121]. A significant decrease was noted in the post-treatment concentrations of MDA and RADAI score.
- An improvement in the antioxidant parameters along with a decline in MDA concentrations after treatment, revealed a negative correlation between MDA and total thiol concentrations and also between MDA and glutathione concentrations. Observations also suggest that the patients receiving antioxidant supplemented drug regimen resulted in prompt recovery of RA symptoms, therefore, suggesting a potential role for antioxidants in lowering the oxidative stress and the resultant inflammatory damage in patients of RA [122]. However, these potential benefits need to be quantified and balanced against well-described toxicity of mega-doses of vitamins.

Pulsed electromagnetic field (PEMF)

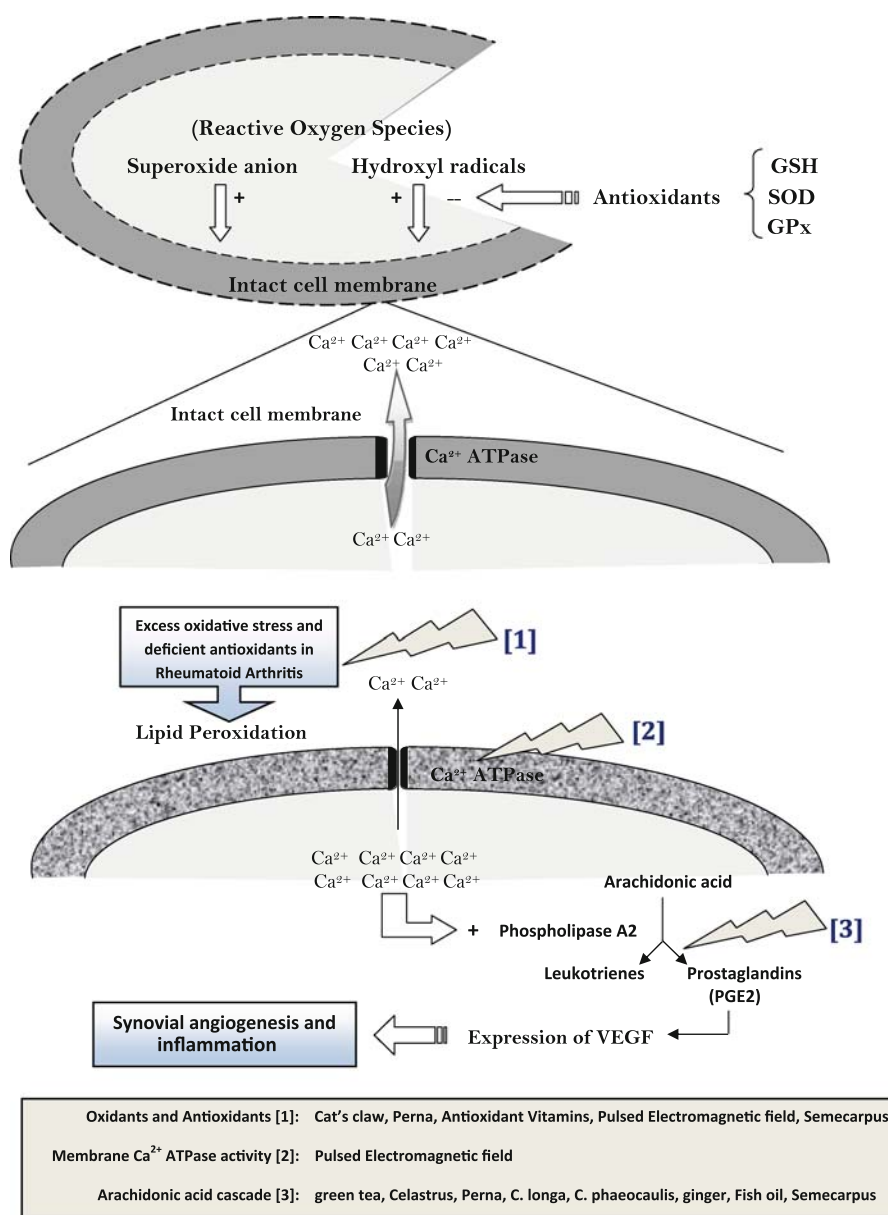
Electromagnetic field implies a safe, non-invasive, and easily applicable therapeutic method for treating painful and inflammatory conditions. Studies over the past two decades

have shown long-term records of safety with the therapeutic use of low-frequency pulsed electromagnetic field (PEMF) [123]. The anti-inflammatory effects of PEMF have been demonstrated in several animal arthritic models [123, 124].

- A study performed using adjuvant-induced arthritis (AIA) models, have shown that low-frequency and low-intensity pulsed electromagnetic field (PEMF) exerts anti-inflammatory effects by restoration of the plasma membrane Ca^{2+} ATPase activity [123]. An uncontrolled production of oxygen-free radicals in the articular cartilage of RA patients can lead to lipid peroxidation, but the antioxidant enzymes viz, glutathione (GSH), superoxide dismutase

(SOD), and glutathione peroxidase (GPx) protect against the oxygen toxicity and lipid peroxidation (Fig. 2). Lipid peroxidation results in an alteration in the fluidity and permeability of plasma membrane, thus, inhibiting the plasma membrane Ca^{2+} ATPase (PMCA) activity, which is responsible for the expulsion of Ca^{2+} from the cytosol of the cells and keeping the cytoplasmic concentration of Ca^{2+} to extremely low levels. As proposed by Almaden et al. [125] an increased concentration of intracellular Ca^{2+} levels may activate phospholipase A2, thereby resulting in release of arachidonic acid and production of PGE2, which is an important mediator of synovial angiogenesis and joint inflammation in RA.

Fig. 2 Targets of CAM modalities in intracellular signaling mechanisms



- Research has shown that low-frequency and low-intensity PEMF could stabilize the PMCA activity, subsequently restoring the intracellular Ca^{2+} concentration to extremely low levels and suppressing the production of proinflammatory PGE2 [123]. Another significant effect of PEMF found in the cells undergoing oxidative stress, was attenuation of the oxidative damage and reverting the antioxidant defense system back to normal.
- Segal et al. [126] performed a double blind clinical trial (Table 3) to assess the efficacy of electromagnetic field in RA patients using two configurations of static magnetic fields. Results revealed that both devices demonstrated statistically significant pain reduction in comparison to baseline values, after 1 week of electromagnetic therapy ($p < 0.01$). Other assessment measures that reached statistically significant values include, subject's global assessment of disease activity and subject's functional assessment of treatment outcome ($p < 0.01$).
- Hence PEMF could prove to be a potential safe, non-invasive, and easily applicable adjunct treatment, if supported by further studies in the field.

Discussion

- With its growing popularity among consumers and professionals worldwide and the emergence of quantitative research using valid clinical end-points, CAM is becoming less “alternative” with each passing year. Lack of knowledge regarding its clinical efficacy and its mechanisms of action does not justify the a priori disapproval of CAM by modern medicine. The increased interest of RA patients in CAM modalities for pain relief has brought up valid questions of safety and efficacy. Reports have shown that more than 70% of patients using CAM never mentioned the use of these products to their physicians. Although generally thought to be safe, some of these potent modalities have been associated with rare but very serious side effects and drug interactions. These adverse effects may be overlooked if the physicians are unaware of the safety profile and mechanisms of action of widely used CAM modalities. In this review we have attempted to build a framework of the proposed mechanisms of action and evaluate the available evidence for the efficacy of widely used CAM modalities in RA. This framework may facilitate further questions and studies that will help enhance our scientific understanding of CAM modalities.
- There is a publication bias that has resulted in a plethora of published studies showing CAM modalities having beneficial effects in RA. However, most of them were based on animal models of RA, and claim an insufficient evidence for the efficacy of CAM modalities in rheumatology rehabilitation. Furthermore, for the great majority of these modalities no type-A evidence (randomized controlled studies) data are available.
- CAM modalities, such as green tea, ginger, green-lipped mussel, and antioxidant vitamins have been commercially popularized for the treatment of RA, despite lack of empirical evidence for their efficacy through clinical trials.
- Only five of the above modalities, acupuncture, BVA, omega-3 PUFAs (fish oil), pulsed electromagnetic field, and herbal extracts of *Uncaria tomentosa* (cat's claw), *Tripterygium wilfordii* hook F and *Curcuma longa* (turmeric) have a substantial evidence of evaluation through RCTs. The remaining modalities reviewed have been evaluated using adjuvant arthritic models. Even within those CAM modalities that have amassed a number of empirical reports through RCTs, the quality of the methodology across studies varied a great deal. Among the clinical studies evaluated for this review, the maximum patients enrolled for a study were 97, whereas most of the studies recruited a sample of even less than 43 patients (Table 3). The small sample size suggest that the studies were not powered enough to support the efficacy of CAM modalities. High dropout rates, either due to lack of efficacy or adverse effects, have been associated with these studies [72, 113]. These methodological limitations need to be addressed in the design of potential future clinical trials, in order to avoid the pitfalls of the past.
- In summary, the lack of high-quality clinical investigations does not permit definitive conclusions to be drawn regarding the efficacy of CAM modalities in RA. However, several animal studies have helped elucidate the potential mechanism of action of various CAM modalities, which may encourage a more careful testing.
- Moreover, the use of CAM modalities is not always without concerns. The potential for interaction between herbal substances and prescription drugs is of concern because the pharmacokinetic characteristics of one or more medications taken simultaneously can be altered and can lead to clinically significant toxicity. Many interactions and contraindications have been documented for herbal preparations. For example, one of the most common interactions includes patients taking warfarin or aspirin who are also taking cod liver oil, evening primrose oil, ginkgo biloba, or garlic [127]. All of these natural remedies have coumarinic constituents and can increase the risk of bleeding. Therefore, there can be significant safety problems with their concomitant use, often exacerbated by the false general belief that these substances are always benign and the patients' hesitance to report their usage to their physicians.

Therefore, while the use of CAM modalities for the treatment of RA continues to increase, rigorous controlled clinical trials examining their efficacy are necessary to validate or refute the clinical claims made for CAM modalities.

Conflict of interest statement The authors have no conflicts of interest.

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